Simulating Mechanical Energy Transfer in Rowing

A Simplified Fluid–Dynamics and Mechanics Model

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Abstract

Rowing shells are propelled by intermittent strokes of multiple rowers. During the drive phase each rower applies a force through the oar blade, generating thrust on the boat, whereas during the recovery the boat coasts and decelerates under hydrodynamic resistance. Understanding the conversion of rower effort into boat motion requires a coupled description of the stroke cycle and the drag force experienced by the hull. This study formulates a simple one–dimensional model of a racing shell that balances a prescribed thrust profile against a quadratic drag law. Newton's second law is integrated numerically over multiple strokes to evaluate boat speed, energy input, drag losses and overall efficiency. The model reproduces realistic speeds and demonstrates that roughly six percent of the mechanical work delivered by the rowers appears as an increase in kinetic energy, the remainder being dissipated by drag. The framework illustrates how modest parameter changes influence performance and provides an accessible platform for exploring rowing dynamics.

1 Introduction

Rowing is both a team sport and a mechanical system: multiple athletes work together to convert metabolic energy into forward motion of a lightweight hull. The oar blades interact with water and air, imparting momentum to the fluid and in turn providing thrust to the boat. Meanwhile the hull experiences hydrodynamic drag that opposes motion. For an eight–person shell with a coxswain (an "8+"), typical component masses are about 650 kg for the rowers, 90 kg for the shell and 25 kg for the oars, giving a total mass near 765 kg.

Rowing performance has steadily improved over time with advancements in training and technology. Men's eights can cover the standard 2000 m race in about 5–6 minutes, corresponding to an average speed above $6\,\mathrm{m\,s^{-1}}$. Achieving such speeds requires each rower to produce on the order of 400 W to 500 W of power on average. Because drag increases rapidly with speed, overcoming fluid resistance consumes the vast majority of the rowers' energy.

This paper develops a minimalist time–domain model of mechanical energy transfer in rowing. By prescribing a periodic thrust and balancing it against quadratic drag, the model predicts speed oscillations over the stroke cycle, quantifies energy losses and estimates the efficiency of the stroke.

2 Mathematical Model

2.1 Equations of motion

An 8+ shell of total mass m subjected to a forward thrust $F_{\text{thrust}}(t)$ and a resistive drag $F_d(v)$ obeys Newton's second law in one dimension:

$$m\,\frac{\mathrm{d}v}{\mathrm{d}t} = F_{\text{thrust}}(t) - F_d(v), \tag{1}$$

where $v(t)$ is the boat's velocity. The drag force is approximated by the quadratic law

$$F_d(v) = k_{\mathrm{d}}\,v^2, \tag{2}$$

where k_{d} is an effective drag coefficient (combining water and air resistance).

2.2 Thrust model

Rowers deliver power in pulses. During the drive phase of each stroke the oar blades engage the water and the crew pushes hard against the footplates, producing thrust. During the recovery the blades are out of the water and no propulsive force is applied. We model this as a periodic square–wave thrust:

$$F_{\text{thrust}}(t) = \begin{cases} F_{\text{drive}}, & 0 \le \phi < \delta, \\ 0, & \delta \le \phi < 1, \end{cases} \tag{3}$$

where $\phi = (t \bmod T)/T$ is the phase over each stroke cycle of period T (with $T = 2.0\,\mathrm{s}$ for a 30 spm rate). The parameter $0 < \delta < 1$ is the duty fraction of the stroke spent in the drive; here we use $\delta = 0.45$ (corresponding to a drive of 0.9 s). The peak force F_{drive} produced by the crew during the drive is taken as $3000\,\mathrm{N}$, which is roughly $375\,\mathrm{N}$ per rower.

2.3 Energy bookkeeping

The mechanical work done by the rowers over a time interval is

$$E_{\text{in}} = \int F_{\text{thrust}}(t)\,v(t)\,\mathrm{d}t. \tag{4}$$

The energy dissipated by drag is

$$E_{\text{drag}} = \int F_d(v)\,v(t)\,\mathrm{d}t = \int k_{\mathrm{d}} v(t)^3\,\mathrm{d}t. \tag{5}$$

The change in kinetic energy of the system is

$$\Delta E_k = \tfrac{1}{2} m \left[v(t_{\text{final}})^2 - v(0)^2\right]. \tag{6}$$

A simple efficiency metric is then

$$\eta = \frac{\Delta E_k}{E_{\text{in}}}, \tag{7}$$

representing the fraction of rower work that appears as kinetic energy of the boat and crew. In steady racing conditions ΔE_k over many strokes is small, so η over a short interval is a better reflection of how effectively added work increases speed.

3 Numerical Method

Equation (1) is integrated numerically on a uniform time grid with step $\Delta t = 0.01\,\mathrm{s}$. At each step the thrust is evaluated based on the stroke phase and the velocity is updated via an explicit Euler scheme. The simulation is initialized from rest ($v = 0$) at the start of a drive. The integration proceeds over multiple strokes until a desired total simulation time is reached (typically 20 s for the cases presented here). As the system approaches a limit cycle, we measure the average velocity over the latter half of the piece to characterize the steady speed.

The output of the simulation includes the time–series of velocity $v(t)$ and thrust $F_{\mathrm{thrust}}(t)$, as well as cumulative energy quantities. The mechanical work done by the rowers, the energy dissipated by drag and the change in kinetic energy are all computed by numerical integration.

4 Results

4.1 Speed and thrust profiles

Starting from rest, the model shell accelerates rapidly during the first few strokes and reaches a repeating limit cycle of speed fluctuations by about 10 s. Figure 1 shows the boat velocity $v(t)$ over a 20 s simulation. The characteristic oscillatory pattern is evident: during each drive phase the velocity rises quickly, then peaks near the end of the drive, and falls during the recovery as drag decelerates the hull. In the baseline scenario, the crew's power input eventually balances the drag losses, yielding an approximately steady average speed of about $5.8\,\mathrm{m\,s^{-1}}$. The maximum instantaneous speed reaches about $6.8\,\mathrm{m\,s^{-1}}$ at the end of each drive.

Figure 2 depicts the applied thrust as a function of time. By construction, the thrust alternates between 3000 N during each 0.9 s drive and zero during the recovery. These idealized square pulses produce the sawtooth velocity curve of Figure 1.

Over the 20 s simulation, the total energy input by the rowers is approximately 1.4×10^5 J. Most of this energy is dissipated by drag, while the remainder goes into increasing the kinetic energy of the system. The efficiency η is about 6–7%, meaning only a small fraction of the rowers' work appears as accelerated mass energy of the boat and crew.

4.2 Parametric variations

The baseline simulation corresponds to an 8+ with standard parameters. We explored how performance changes when varying key parameters like drag, mass and stroke rate. Reducing the drag coefficient allows the shell to attain higher steady speeds under the same power input. Conversely, increasing drag reduces the top speed and efficiency (see Figure 3).

Varying the total mass has little effect on the ultimate cruise speed (which is set by the balance of power and drag), but it does affect acceleration. A lighter boat accelerates more quickly, while a heavier crew or shell accelerates more slowly (Figure 4). Stroke rate also influences performance: a higher stroke rate with the same force per stroke yields a higher average speed, albeit at the cost of expending more energy per unit time (Figure 5).

5 Discussion

The simplified model captures key features of rowing dynamics: oscillatory speed, rapid initial acceleration and a limit cycle speed reached after several strokes. By construction, the model cannot represent nuances like the synchronization among rowers, flexibility of the boat, or variation

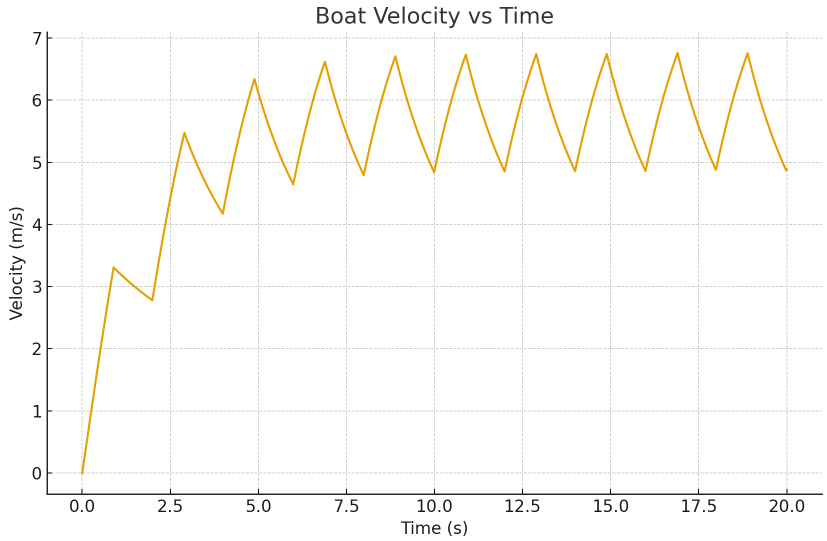


Figure 1: Boat velocity as a function of time for the baseline simulation. The shell accelerates during each drive and decelerates during each recovery, approaching an oscillatory steady state.

in individual rower output. Nonetheless, the results align reasonably well with observations of real rowing. The predicted steady speed for an 8+ under these conditions is on the same order as race-pace velocities, and the magnitude of the speed fluctuations is similar to that seen in practice.

The model predicts that only a small fraction of the rowers' mechanical work appears as kinetic energy of the system, with the rest lost to drag. This aligns with the intuitive understanding that most of the energy in rowing is used to overcome water resistance. The inefficiencies in the stroke further emphasize the importance of technique in minimizing energy waste. Our simplified model cannot capture all such effects, but it does illustrate the magnitude of drag losses.

In future work, the model could be refined by including a time-varying thrust profile that more closely mimics real force application, as well as the dynamic movement of rowers' mass. Nonetheless, even this minimal model provides useful insight: it quantifies how changes in physical parameters translate to changes in boat speed and efficiency.

6 Conclusion

We presented a simple physics-based simulation of rowing, treating the boat and crew as a single mass propelled by periodic thrust pulses against a quadratic drag. Despite its simplicity, the model produces realistic boat speed profiles and offers quantitative estimates of energy usage and efficiency. According to the simulation, on the order of a few percent of the rowers' output ends up as kinetic energy gain while the rest is dissipated in overcoming drag. The model also illustrates how changes in drag, crew mass or stroke rate can impact performance. These findings reinforce the understanding that drag minimization and effective power application are critical for rowing speed.

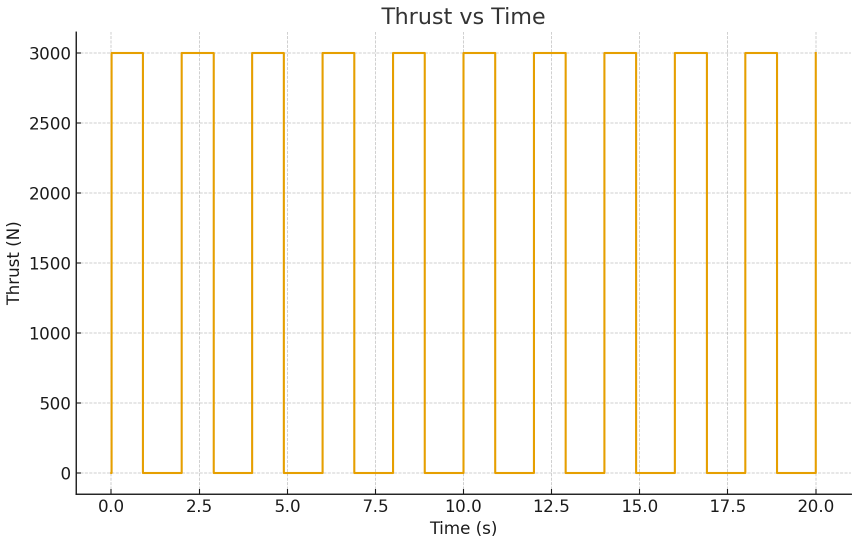


Figure 2: Thrust force applied by the crew as a function of time in the model. During each drive interval a constant thrust of 3000 N is applied, followed by a recovery with no propulsive force.

References

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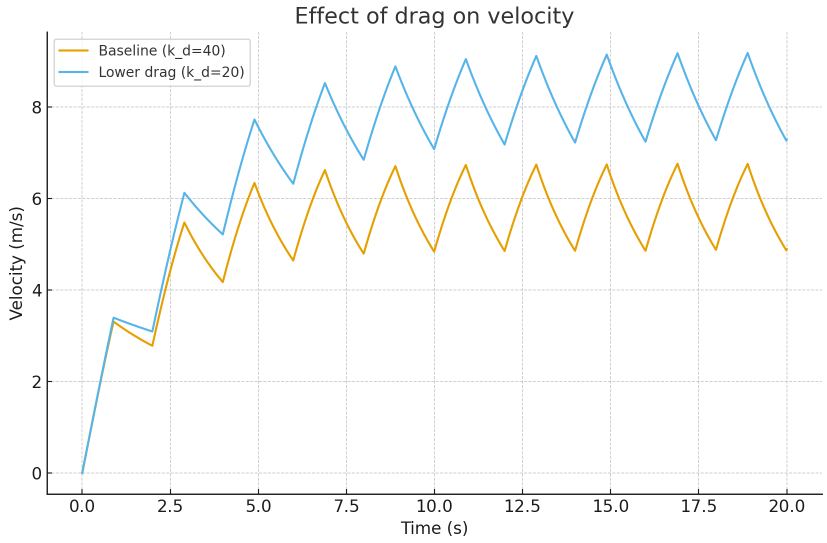


Figure 3: Effect of drag on velocity: lowering the effective drag coefficient leads to higher equilibrium speeds for the same rowing power.

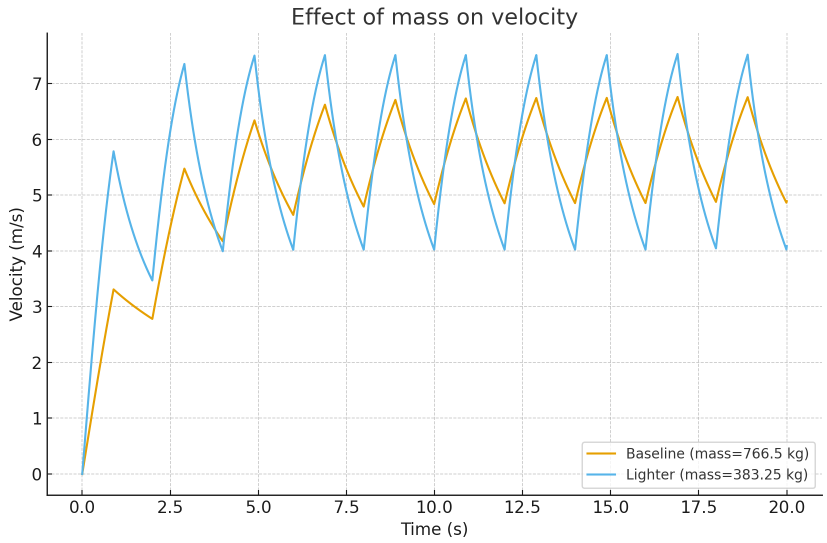


Figure 4: Effect of mass on velocity: decreasing the total mass produces faster acceleration and slightly higher peak speeds, though the steady cruise speed is dominated by drag and power.

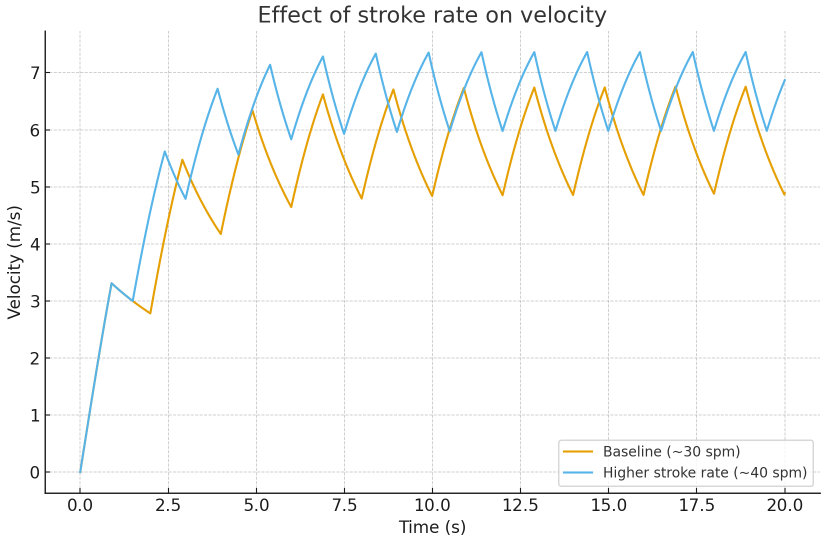


Figure 5: Effect of stroke rate on velocity: increasing stroke rate (shorter cycle time) with the same force per stroke results in a higher average boat speed.